Session 8: Simple Linear Regression

Context & Objective

Textbook: Chapter 14 – Simple Linear Regression & Correlation

Objective: Students build a simple regression model to evaluate the relationship between retail store floor space and sales revenue, checking slope significance and using predictions for business decisions.

1. Antigravity Skill Definition (simple-regression-modeler)

Skill Name: simple-regression-modeler

Purpose: Computes simple linear regression statistics for one independent variable (X) and one dependent variable (Y). Calculates the regression equation, goodness-of-fit metrics, tests slope significance, and produces predictions with confidence intervals.

1.1 Core Logic & Formulas

Step 1: Input Paired Data (X, Y)

X = Independent variable (predictor) – in this case, store floor space in square feet

Y = Dependent variable (response) – in this case, weekly sales revenue in thousands of dollars

Step 2: Calculate Least Squares Estimates

Calculate means: $\bar{x}$ = mean of X values, $\bar{y}$ = mean of Y values

Calculate $S_{xx} = \Sigma(x_i - \bar{x})^2$ and $S_{xy} = \Sigma(x_i - \bar{x})(y_i - \bar{y})$

Slope: $b_1 = S_{xy} / S_{xx}$ (change in Y per unit increase in X)

Intercept: $b_0 = \bar{y} - b_1 \times \bar{x}$ (predicted Y when X = 0)

Step 3: Compute Sum of Squares & R-Squared

SST (Total) = $\Sigma(y_i - \bar{y})^2$ – total variation in Y

SSR (Regression) = $b_1^2 \times S_{xx}$ – variation explained by X

SSE (Error/Residual) = SST - SSR – unexplained variation

R^2 = SSR / SST – proportion of Y variation explained by X (0 to 1)

Step 4: Compute Standard Errors

Standard Error of Estimate: $s = \sqrt{(SSE / (n-2))}$ – average distance from regression line

Standard Error of Slope: s_{b1} = $s / \sqrt{(S_{xx})}$ – precision of slope estimate

Step 5: Hypothesis Test for Slope Significance

t-statistic = b_1 / s_{b1} with degrees of freedom = n - 2

H_0: $\beta_1 = 0$ (X does not predict Y) vs H_1: $\beta_1 \neq 0$ (X predicts Y)

If p-value < α or |t| > t_critical: Reject H_0 (slope is statistically significant)

2. Student Execution Prompt

Model the relationship between retail store floor space (X in square feet) and weekly sales revenue (Y in thousands of dollars):

Independent Variable (X – Floor Space): [100, 150, 200, 250, 300] sq ft

Dependent Variable (Y – Sales): [20, 28, 32, 41, 49] (in $1000s)

Task: Calculate the regression equation $\hat{y} = b_0 + b_1 \times X$. Compute R-squared, standard error of b_1, and t-statistic for the slope at significance level $\alpha = 0.05$. Interpret the results and make a prediction for a 100 sq ft store expansion.

Raw Data & Descriptive Statistics:

Store ID	Floor Space (X)	Weekly Sales (Y)
Store 1	100 sq ft	$20k
Store 2	150 sq ft	$28k
Store 3	200 sq ft	$32k
Store 4	250 sq ft	$41k
Store 5	300 sq ft	$49k

3. Managerial Decision Output

Regression Model Results:

Regression Equation: $\hat{y} = 6.4 + 0.138 \times$ (Floor Space in sq ft)

Intercept (b_0): $6,400 (predicted sales with zero floor space)

Slope (b_1): 0.138 (each additional sq ft generates $138 in weekly sales)

Model Fit & Significance:

R-Squared: 98.4% (floor space explains 98.4% of sales variation)

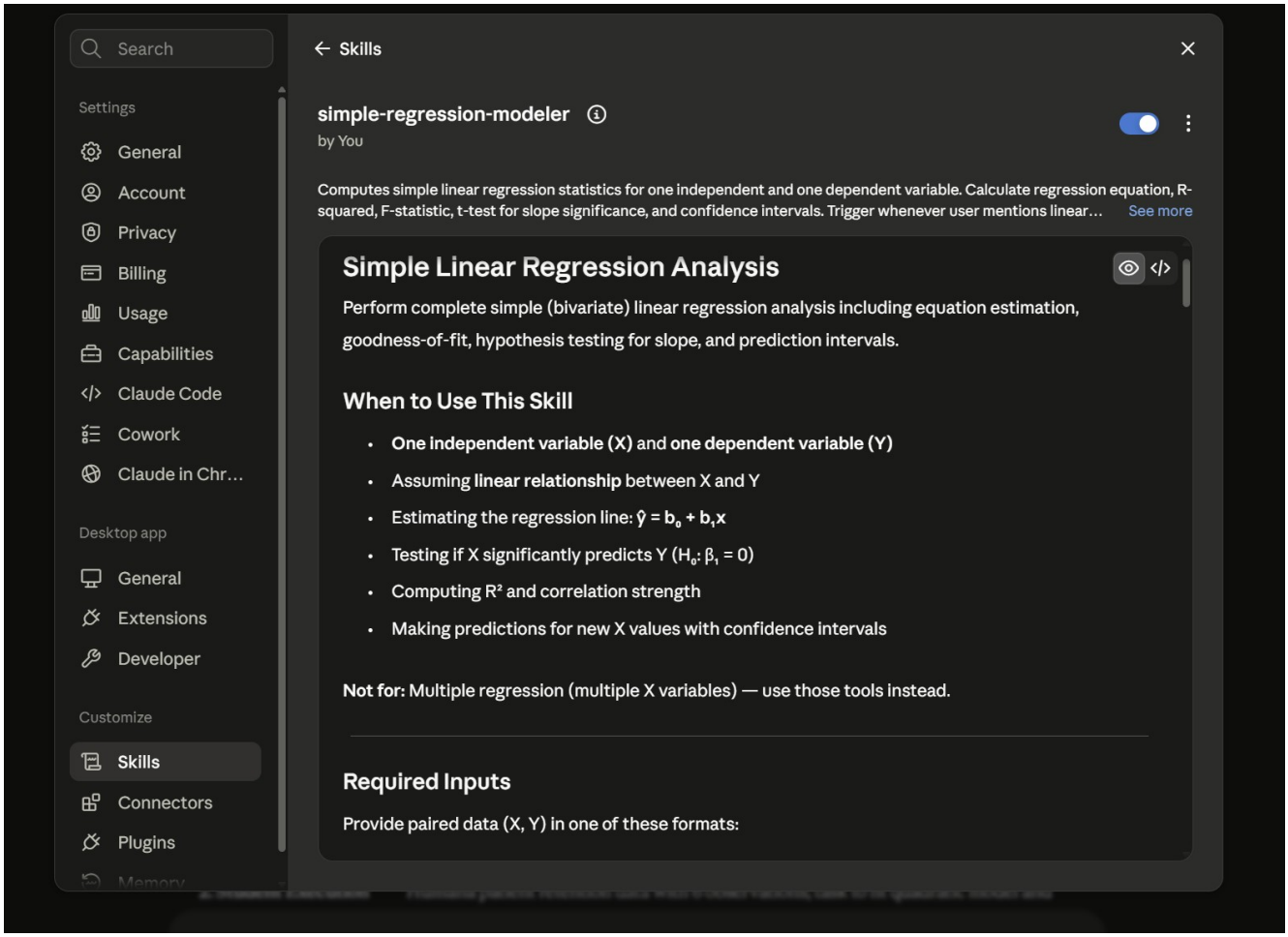
Standard Error of Slope (s_{b1}): 0.0101

t-Statistic: 13.6 (df = 3)

p-Value: < 0.001 (highly statistically significant)

Managerial Decision:

The store planner's regression analysis demonstrates an exceptionally strong, linear relationship between floor space and weekly sales revenue ($R^2 = 98.4\%$, $t = 13.6$, $p < 0.001$). The model is highly significant, with floor space explaining virtually all variation in sales. The slope coefficient of 0.138 indicates that each additional square foot of retail space generates approximately $138 in weekly sales revenue. Given this compelling relationship, the planner approves a proposed 100 square foot store expansion. Based on the regression model, this expansion is projected to increase weekly sales by $13,800 ($0.138 \times 100 =$ $13.8k), a substantial revenue lift with clear ROI implications. The planner recommends moving forward with the expansion project.


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simple-regression-modeler
by You
Computes simple linear regression statistics for one independent and one dependent variable. Calculate regression equation, R-squared, F-statistic, t-test for slope significance, and confidence intervals. Trigger whenever user mentions linear… See more
Simple Linear Regression Analysis
Perform complete simple (bivariate) linear regression analysis including equation estimation, goodness-of-fit, hypothesis testing for slope, and prediction intervals.
When to Use This Skill
One independent variable (X) and one dependent variable (Y)
Assuming linear relationship between X and Y
Estimating the regression line: ŷ = b₀ + b₁x
Testing if X significantly predicts Y (H₀: β₁ = 0)
Computing R² and correlation strength
Making predictions for new X values with confidence intervals
Not for: Multiple regression (multiple X variables) — use those tools instead.
Required Inputs
Provide paired data (X, Y) in one of these formats: